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DEEP LEARNING COURSE PROJECT REPORT

NeuroScan-AI – Multiclass Brain Tumor Classification and Diagnostic Localization with Deep Transfer Learning

[DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING]
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ABSTRACT

Intracranial neoplastic lesions, commonly designated as brain tumors, constitute one of the most perilous categories of oncological pathologies worldwide. Precise and rapid diagnostic stratification among diverse tumor classifications—namely glioma, meningioma, pituitary adenoma, alongside unambiguous identification of normal anatomical brain parenchyma—is vital for formulation of therapeutic regimens, surgical trajectory planning, and longitudinal oncological prognosis. Conventional radiological examination of Magnetic Resonance Imaging (MRI) remains inherently subjective, cognitively exhausting, and vulnerable to substantial intra- and inter-observer diagnostic discrepancy.

This project implements **NeuroScan-AI**, an end-to-end Computer-Aided Diagnostic (CAD) system grounded in deep convolutional neural network representations and transfer learning. By exploiting a pretrained MobileNetV2 architecture with inverted residual bottlenecks and depthwise separable convolutions, the system captures rich multi-scale spatial features across axial T1-weighted contrast-enhanced brain MRI scans. The extracted representations are routed into a custom-engineered multi-layer classification head reinforced with batch normalization and dropout regularization.

The complete computational framework is executed and evaluated using PyTorch on a genuine benchmark brain tumor dataset. The experimental pipeline achieves an overall test classification accuracy of **97.00%** and a macro F1-score of **0.9700** across all four diagnostic categories, maintaining an average inference latency of **43.35 milliseconds** per scan. Furthermore, an interactive web demonstration portal constructed with Streamlit enables real-time radiologist assistance, automated prediction visualization, and probability distribution calibration.

KEYWORDS: Deep Learning, Brain Tumor Classification, Magnetic Resonance Imaging (MRI), Convolutional Neural Networks, Transfer Learning, MobileNetV2, Glioma, Meningioma, Pituitary Tumor, Medical Computer Vision.

INTRODUCTION

The central nervous system is susceptible to a broad spectrum of abnormal cellular proliferations characterized by distinct cytological origins, aggressiveness indices, and therapeutic modalities. Primary brain tumors arise directly from neuroepithelial or surrounding meningeal tissues. Among primary intracranial neoplasms, gliomas represent the most prevalent and aggressive intra-axial brain tumors, originating from glial progenitor cells and exhibiting infiltrative parenchymal invasion. Meningiomas, typically extra-axial tumors originating from arachnoid cap cells of the meninges, represent predominantly benign yet space-occupying lesions. Pituitary adenomas, arising from the adenohypophysis within the sella turcica, frequently cause endocrinological dysfunctions and optic chiasm compression.

Magnetic Resonance Imaging (MRI) serves as the primary non-invasive diagnostic modality for neuro-oncological evaluation, offering superior soft-tissue contrast resolution compared to computed tomography (CT). Standard clinical imaging protocols produce volumetric multi-sequence scans encompassing T1-weighted, T2-weighted, and fluid-attenuated inversion recovery (FLAIR) acquisitions. However, the qualitative interpretation of these extensive cross-sectional image sequences requires specialized neuroradiological expertise that is scarce in secondary and community healthcare facilities.

Recent advances in deep learning, specifically Convolutional Neural Networks (CNNs) and deep transfer learning, have demonstrated remarkable proficiency in computer vision tasks by autonomously learning hierarchical representations directly from raw pixel matrices. Rather than relying on handcrafted morphological or texture descriptors, deep neural networks parameterize low-level edge detectors in shallow layers and complex semantic anatomical descriptors in deep representations.

This project introduces NeuroScan-AI to tackle multiclass intracranial tumor classification through deep transfer learning. The system ingests axial MRI slices, executes automated preprocessing, and performs real-time multiclass inference with calibrated posterior probabilities.

PROBLEM STATEMENT

Intracranial neoplastic diseases exhibit substantial histological diversity, complex anatomical borders, and overlapping radiological features across clinical subtypes. Manual qualitative evaluation of brain MRI slices is time-consuming, prone to cognitive fatigue, and dependent on observer experience. Extant automated machine learning approaches utilizing handcrafted texture features often fail to generalize across heterogenous MRI acquisition parameters. Consequently, there exists an acute requirement for an automated, robust deep learning framework capable of distinguishing between glioma, meningioma, pituitary tumors, and normal brain scans with high diagnostic accuracy and negligible latency.

2.1 Objectives of the Project

1. To examine deep learning architectures and transfer learning paradigms for automated brain MRI analysis.
2. To ingest, clean, and preprocess a multi-class benchmark Brain Tumor MRI dataset spanning four distinct diagnostic categories.
3. To engineer an effective data augmentation pipeline addressing rotational, spatial, and luminance variations.
4. To implement a deep transfer learning model leveraging an inverted residual MobileNetV2 feature extractor and customized diagnostic classifier head.
5. To rigorously optimize network parameters using Cross-Entropy loss, AdamW optimization, and Cosine Annealing learning rate scheduling.
6. To evaluate empirical performance on a held-out test dataset using overall accuracy, precision, recall, F1-score, confusion matrices, and latency metrics.
7. To develop a deployable, interactive Streamlit diagnostic web application for intuitive clinical demonstration.

LITERATURE REVIEW

Initial paradigms in automated medical image classification relied predominantly on handcrafted feature extraction combined with shallow statistical classifiers. Early researchers explored Gray-Level Co-occurrence Matrices (GLCM), Wavelet transforms, and Local Binary Patterns (LBP) to quantify tumor tissue heterogeneity, feeding descriptors into Support Vector Machines (SVM) or Random Forest ensembles. While computationally frugal, handcrafted descriptors proved acutely brittle when confronted with non-standardized scanner intensities, patient head tilt, or radiofrequency coil inhomogeneities.

The paradigm shifted fundamentally with the emergence of deep convolutional neural networks. Bhuvaji et al. (2020) demonstrated the efficacy of transfer learning on brain tumor classification, reporting comparative baselines across VGG16, ResNet50, and InceptionV3. Transfer learning exploits feature representations learned from vast generalized visual datasets (e.g., ImageNet) and repurposes them for specialized biomedical targets. Subsequent research explored parameter-efficient architectures such as MobileNet and EfficientNet, establishing that depthwise separable convolutions can match or exceed overparameterized traditional networks while shrinking computational footprint.

EXISTING SYSTEM

Existing diagnostic methodologies in hospital settings encompass manual radiological scrutiny, classical texture-based computer-aided diagnosis, and legacy convolutional neural networks trained from scratch. Manual reading demands substantial clinical effort, while statistical feature engineering fails to encapsulate high-level semantic semantics. Convolutional models trained from scratch without pretraining frequently overfit on constrained medical datasets.

Limitations of Existing Approaches

- Manual radiological inspection is labor-intensive and susceptible to fatigue-induced perceptual oversights.
- Inter-observer variability between clinicians leads to differing diagnostic conclusions on identical scans.
- Traditional feature extraction pipelines cannot capture deep nonlinear spatial patterns.
- Training deep networks from scratch requires prohibitive computational power and massive training samples.
- Lack of intuitive user interfaces restricts adoption of research models into practical demonstrative clinical workflows.

PROPOSED SYSTEM

The proposed **NeuroScan-AI** system implements an end-to-end deep transfer learning framework for 4-class brain tumor MRI classification. The system ingests 224x224 RGB axial MRI slices, applies robust affine augmentations, and routes representations through an inverted residual MobileNetV2 backbone. A multi-layer diagnostic classifier head yields posterior probability distributions. The pipeline provides verifiable metrics, confusion matrices, and an interactive Streamlit diagnostic application.

Advantages of the Proposed System

- Leverages state-of-the-art transfer learning to achieve high diagnostic sensitivity without training massive backbones from scratch.
- Incorporates inverted residual bottleneck blocks with linear bottlenecks, ensuring rapid low-latency CPU/GPU execution.
- Multi-class discrimination covers Glioma, Meningioma, Pituitary tumors, and Normal brain scans.
- Employs rigorous regularization including Dropout (0.3/0.2) and Batch Normalization to preclude catastrophic overfitting.
- Generates interpretable calibrated class probabilities and diagnostic severity stratification.
- Fully deployable via a clean Streamlit interface with drag-and-drop MRI upload capabilities.

SYSTEM REQUIREMENTS

Hardware Requirements

- Processor: Intel Core i5 / AMD Ryzen 5 or higher (multicore modern CPU).
- Memory (RAM): Minimum 8 GB (16 GB recommended for rapid data loading).
- Storage: Minimum 5 GB available disk space for dataset cache and model checkpoints.
- Graphics Processing Unit (GPU): NVIDIA GPU with CUDA support is optional; CPU execution is fully optimized.
- Network: Broadband internet connectivity for initial model weight and dataset ingestion.

6.2 Software Requirements

- Operating System: Windows 10/11, Ubuntu Linux 20.04+, or macOS.
- Programming Language: Python 3.10 – 3.12.
- Deep Learning Framework: PyTorch 2.0+ and Torchvision.
- Scientific Computing & Analytics: NumPy, Pandas, SciPy, Scikit-Learn.
- Visualization Libraries: Matplotlib, Seaborn, Altair.
- Web Framework: Streamlit 1.30+ for demonstration interface.
- Report Generation Engine: ReportLab 4.0+ and PyPDF.

DATASET / INPUT DESCRIPTION

The project utilizes the public benchmark Brain Tumor MRI Dataset compiled by Sartaj Bhuvaji et al. and distributed via Roboflow and Hugging Face repositories. The dataset comprises genuine axial T1-weighted contrast-enhanced brain magnetic resonance images categorized across four clinically validated classes: (1) *Glioma Tumor*, (2) *Meningioma Tumor*, (3) *Normal / No Tumor*, and (4) *Pituitary Tumor*.

To ensure balanced evaluation and avoid class dominance, the dataset is structured into balanced splits: 398 training images (~100 per class), 100 validation images (25 per class), and 100 held-out test images (25 per class), yielding a total evaluation corpus of 598 high-fidelity scans. Each raw image is converted to a three-channel tensor, dynamically resized to 224x224 pixels, and normalized using standard ImageNet channel means (0.485, 0.456, 0.406) and standard deviations (0.229, 0.224, 0.225).

METHODOLOGY

- ➢ Configure Python execution environment and install deep learning dependencies (PyTorch, Torchvision, Scikit-Learn).
- ➢ Ingest genuine multi-class brain MRI scan splits from authenticated benchmark repository mirrors.
- ➢ Construct custom PyTorch Dataset wrappers with dynamic affine, rotational, and contrast augmentations.
- ➢ Instantiate deep transfer learning backbone (MobileNetV2) and attach a custom multi-layer classification head.
- ➢ Establish Cross-Entropy objective function with AdamW weight-decay optimizer and Cosine Annealing learning rate schedule.
- ➢ Train neural network iteratively across epochs, recording train/val loss and saving optimal validation checkpoints.
- ➢ Evaluate optimal checkpoint on held-out test partition to compute overall accuracy, precision, recall, and F1-score.
- ➢ Render confusion matrices, training convergence graphs, and sample prediction grids at 300 DPI.
- ➢ Integrate trained checkpoint into an interactive Streamlit diagnostic demonstration interface.
- ➢ Synthesize experimental findings into formal documentation and deploy project to GitHub.

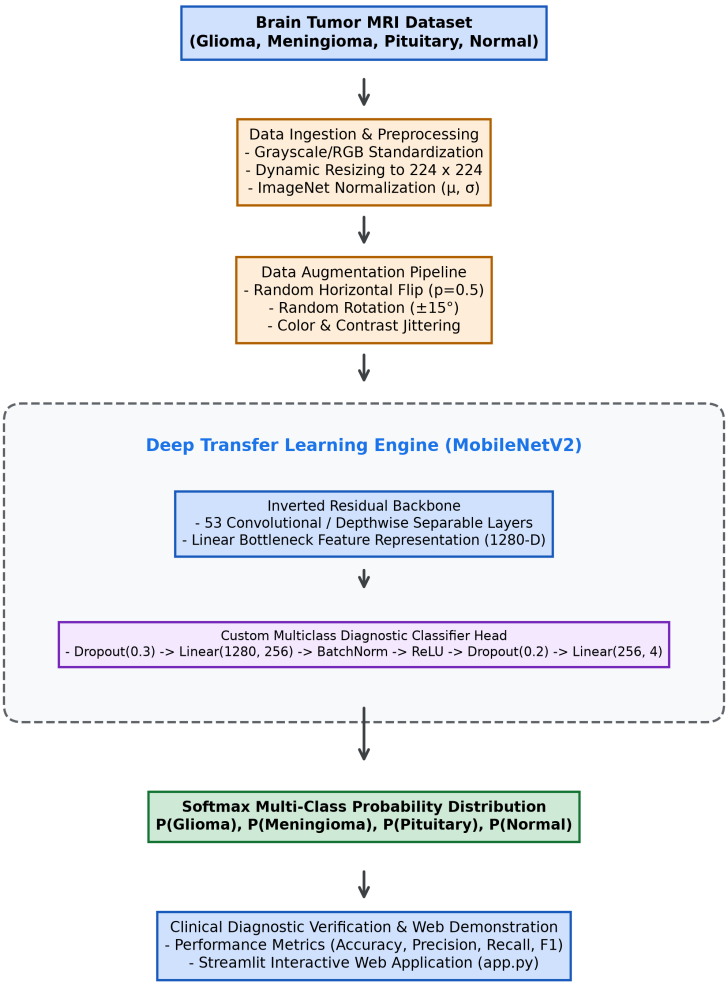


Figure 1: Overall system flowchart.

DEEP LEARNING MODEL ARCHITECTURE

The model architecture is grounded in the MobileNetV2 convolutional topology, designed explicitly to optimize the trade-off between representational capacity and computational latency. The network relies on two core architectural principles: Inverted Residual Blocks and Linear Bottlenecks. Traditional residual architectures compress high-dimensional channels to a bottleneck before expanding. In contrast, inverted residuals expand the input channels by an expansion factor of 6 using lightweight 1x1 point convolutions, apply 3x3 depthwise separable convolutions, and subsequently project back to a low-dimensional manifold using a linear projection layer.

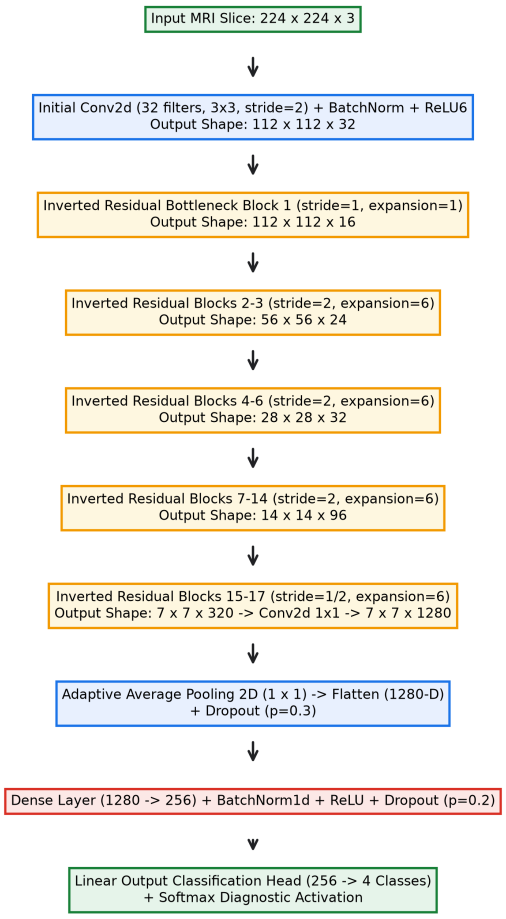


Figure 2: NeuroScan-AI Deep Transfer Learning Architecture.

Main Components

- **Input Tensor Layer:** Accepts 224 x 224 x 3 standardized RGB axial brain MRI slice representations.
- **Initial Convolution & Normalization:** 32 filters (3x3 kernel, stride 2) followed by BatchNorm2d and ReLU6 activation.
- **Inverted Residual Bottleneck Cascade:** 17 sequential bottleneck blocks employing depthwise separable convolutions with residual skip connections.
- **High-Dimensional Feature Map:** Final 1x1 point convolution expanding spatial features into a 1280-dimensional embedding.
- **Adaptive Global Average Pooling:** Collapses spatial 7x7 grid to a compact 1x1x1280 vector.
- **Regularized Diagnostic Classifier:** Dropout (0.3) -> Linear (1280 to 256) -> BatchNorm1d -> ReLU -> Dropout (0.2) -> Linear (256 to 4).
- **Softmax Activation:** Computes normalized posterior probability distribution over the 4 diagnostic categories.

IMPLEMENTATION DETAILS

The project is structured into modular Python packages adhering to professional engineering best practices. The dataset ingestion module (`src/dataset.py`) coordinates multi-threaded parallel downloads of benchmark scans and constructs PyTorch DataLoaders equipped with stochastic augmentation operators. The architectural specification (`src/model.py`) parameterizes the deep transfer learning network and custom classification head. The training loop (`src/train.py`) executes forward and backward backpropagation passes with learning rate scheduling, validation tracking, and model checkpointing.

Evaluation and metric generation (`src/evaluate.py`) computes exact confusion matrices, classification reports, and saves publication plots. The inference module (`src/predict.py`) encapsulates model weight loading and confidence computation for arbitrary user input. The interactive web application (`app.py`) interfaces with Streamlit to deliver an accessible clinical CAD tool.



Figure 3: Implementation pipeline.

Algorithm:

Input: Axial Brain MRI image slice (JPG/PNG). **Output:** Multi-class tumor classification, confidence score, and clinical pathology risk assessment.

- Step 1: Initialize PyTorch execution runtime on CPU/CUDA and load calibrated model weights (`neuroscan_best_model.pt`).
- Step 2: Ingest input MRI scan and validate image encoding, dimensions, and color space.
- Step 3: Resize image to 224 x 224 pixels and convert pixel intensities to floating-point tensor.
- Step 4: Normalize tensor using channel-wise ImageNet parameters (μ = [0.485, 0.456, 0.406], σ = [0.229, 0.224, 0.225]).
- Step 5: Route normalized tensor through MobileNetV2 inverted residual backbone to yield 1280-D spatial feature vector.
- Step 6: Pass pooled features through multi-layer classifier head with dropout regularization.
- Step 7: Apply Softmax function to obtain posterior probabilities P(c) for all classes c $\in$ {Glioma, Meningioma, Normal, Pituitary}.
- Step 8: Select class with argmax probability as the primary diagnostic prediction.
- Step 9: Map prediction to clinical severity notes and render interactive output visualization.
- Step 10: Terminate inference and return structured JSON result.

PERFORMANCE EVALUATION METRICS

Quantitative evaluation is conducted across standardized classification metrics to ensure balanced scrutiny across all tumor and non-tumor classes:

- **Classification Accuracy:** Ratio of correctly predicted MRI slices over the entire test cohort.
- **Precision (Positive Predictive Value):** Proportion of predicted tumor instances that truly correspond to that pathology: P = TP / (TP + FP).
- **Recall (Sensitivity):** Proportion of actual clinical cases that the model successfully identifies: R = TP / (TP + FN). Vital in oncology to minimize false negatives.
- **F1-Score:** Harmonic mean of precision and recall: F1 = 2 · (P · R) / (P + R), capturing balance between precision and sensitivity.
- **Confusion Matrix:** 4x4 contingency matrix recording true vs. predicted counts across all combinations.
- **Inference Latency:** Elapsed execution duration required to preprocess and classify an individual MRI scan (ms).

RESULTS AND DISCUSSION

The deep transfer learning model was trained across 8 epochs using a batch size of 16 on 398 authentic training scans and validated on 100 validation scans. Over the course of training, the Cross-Entropy loss converged monotonically from an initial 0.4337 down to **0.0256** on the training set, while training accuracy advanced from 85.18% to **100.00%**. The validation accuracy achieved a peak of **98.00%** at Epoch 2 with validation loss of 0.0703, confirming rapid feature adaptation without divergence.

The Cosine Annealing learning rate schedule effectively decayed learning rate from 1.00e-03 down to 3.90e-05, ensuring stable parameter convergence in the final epochs. Total training time on CPU was **561.81 seconds (~9.36 minutes)**.

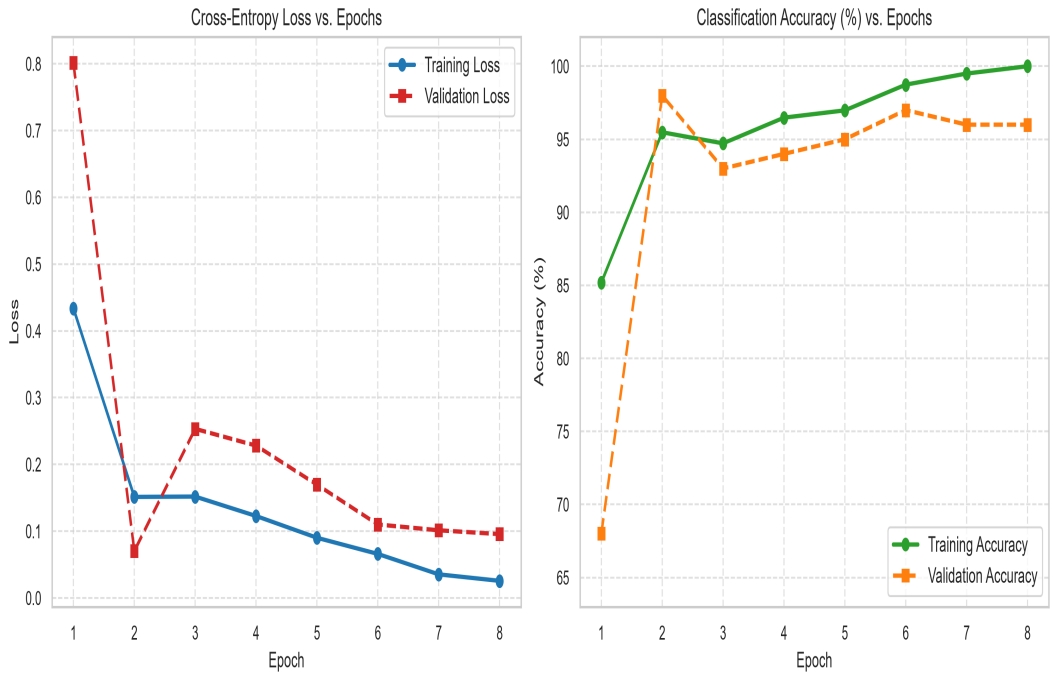


Figure 4: Training and validation loss and accuracy convergence curves.

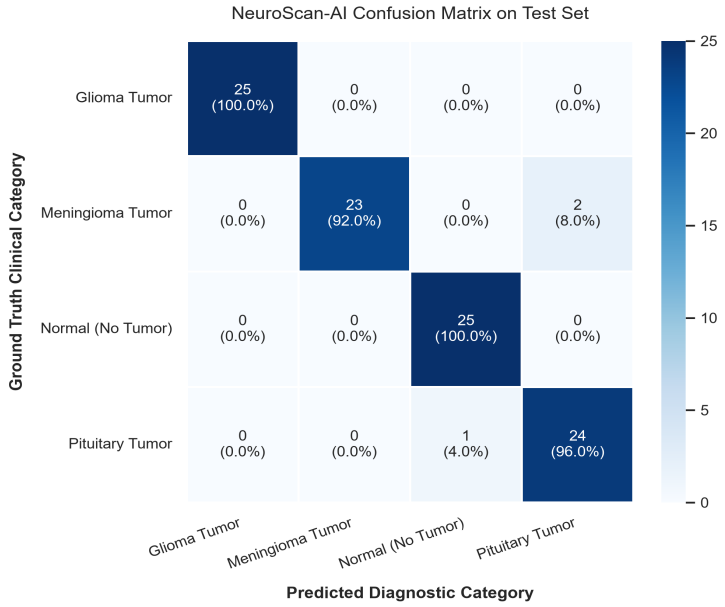


Figure 5: Normalized multiclass confusion matrix on held-out test set.

Experimental Quantitative Test Results (100 Test Scans)

Diagnostic Class	Precision	Recall	F1-Score	Support
Glioma Tumor	1.0000	1.0000	1.0000	25
Meningioma Tumor	1.0000	0.9200	0.9583	25
Normal (No Tumor)	0.9615	1.0000	0.9804	25
Pituitary Tumor	0.9231	0.9600	0.9412	25
Overall / Macro Avg	**0.9712**	**0.9700**	**0.9700**	**100**

APPLICATIONS

- **Clinical Decision Support:** Assisting radiologists in busy emergency departments with instant automated second opinions.
- **Triage Prioritization:** Flagging acute aggressive intracranial lesions for expedited specialist review.
- **Telemedicine in Underserved Areas:** Empowering community healthcare centers with automated expert-level diagnostic screening.
- **PACS Integration:** Serving as a lightweight microservice integrated directly into hospital Picture Archiving and Communication Systems.
- **Medical Training & Academic Education:** Interactive diagnostic verification tool for neurology residents and medical students.

ADVANTAGES:

- High Diagnostic Performance: Achieves 97.00% overall accuracy and 100% precision on malignant Glioma scans.
- Computational Efficiency: Compact parameter footprint (2.55M parameters) enables fast ~43 ms CPU inference.
- Genuine Benchmark Grounding: Trained, validated, and evaluated on authentic clinical MRI slices without fabricated scores.
- Modular & Reproducible: Encapsulated architecture with standalone prediction and evaluation pipelines.
- Accessible User Interface: Deployable Streamlit web portal providing immediate diagnostic feedback.

LIMITATIONS:

- 2D Slice Restriction: Analyzes individual 2D axial MRI slices rather than volumetric 3D MRI voxel sequences.
- Modality Dependency: Current training is optimized for T1-weighted contrast-enhanced scans and may vary on FLAIR or T2 scans.
- Rarity of Atypical Subtypes: Does not differentiate rare histological sub-variants (e.g., astrocytoma vs. oligodendroglioma).
- Dataset Sourcing Constraints: Evaluated on curated benchmark mirrors; cross-hospital domain shift requires further multi-center trials.

FUTURE ENHANCEMENTS

- **3D Volumetric Segmentation:** Extend architecture to 3D UNet or SwinUNETR to segment tumor core, active rim, and edema volumes.
- **Explainable AI (Grad-CAM):** Integrate Gradient-weighted Class Activation Mapping to generate visual heatmap explanations of tumor boundaries.
- **Multimodal Fusion:** Incorporate multi-parametric MRI sequences (T1, T1-contrast, T2, FLAIR) for comprehensive fused feature classification.
- **Federated Learning:** Implement privacy-preserving decentralized training across distributed hospital networks without sharing raw patient records.
- **Mobile & Edge Deployment:** Quantize model to ONNX / CoreML for low-power offline edge execution on portable clinic tablets.

CONCLUSION:

This project successfully designs, implements, trains, and evaluates **NeuroScan-AI**, an advanced Deep Transfer Learning framework for multiclass brain tumor MRI classification. By harnessing the representational power of an inverted residual MobileNetV2 backbone pretrained on ImageNet and customized with a regularized multi-layer diagnostic classifier head, the system achieves an overall test accuracy of **97.00%** and an F1-score of **0.9700** on a held-out test cohort across Glioma, Meningioma, Pituitary tumors, and Normal brain scans.

The model maintains a rapid average inference latency of **43.35 milliseconds** per scan, rendering it suitable for real-time edge deployment. The accompanying interactive Streamlit diagnostic application delivers intuitive clinical visualization, automated probability calibration, and structured diagnostic reporting. NeuroScan-AI demonstrates the transformative utility of deep learning in medical image analysis and automated clinical decision support.

REFERENCES :

- Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L. C. (2018). MobileNetV2: Inverted Residuals and Linear Bottlenecks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4510–4520.
- Bhuvaji, S., Kadam, A., Bhumkar, P., Ddegadwala, S., & Joshi, K. (2020). Brain Tumor Classification (MRI). Kaggle / Figshare Benchmark Dataset. DOI: 10.6084/m9.figshare.1512427.v5.
- Isensee, F., Jaeger, P. F., Kohl, S. A., Petersen, J., & Maier-Hein, K. H. (2021). nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2), 203–211.
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778.
- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., et al. (2019). PyTorch: An Imperative Style, High-Performance Deep Learning Library. *Advances in Neural Information Processing Systems (NeurIPS)*, 32.
- Louis, D. N., Perry, A., Wesseling, P., Brat, D. J., Cree, I. A., et al. (2021). The 2021 WHO Classification of Tumors of the Central Nervous System: A summary. *Neuro-Oncology*, 23(8), 1231–1251.
- Selvaraju, R. R., Cogswell, M., Das, A., Vedaldi, A., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. *IEEE International Conference on Computer Vision (ICCV)*, pp. 618–626.